Displayed Is Not Deliverable: Capacity Certificates and Quote Firmness in Inference Routing

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Abstract

Inference marketplaces expose provider quotes, but a cheap displayed quote need not be executable for a particular account when demand arrives. We study the resulting distinction between displayed and deliverable inference liquidity. Capacity-certified score routing is a capped water-fill that weakly improves delivered count over uncapped score routing under hard commitments. A robust linear program improves the worst-case delivery floor on a declared joint-outage support. For private capacity, cost, and reliability, we give a restricted finite-slot VCG-plus-audit construction and show why finite liability alone cannot elicit continuous reliability. We then introduce a randomized micro-probe design: randomizing policy order identifies first-exposure routing effects despite arbitrary effects on later probes, and success/cost effects identify a value-indexed delegation frontier. We show that a default-versus- pinned experiment alone cannot distinguish fallback option value from hidden selection value, and give a three-arm design that separately identifies them. Because public-provider support selects which model-hours can enter that design, we distinguish the eligible-block estimand from the all-candidate estimand and give design-based bounds for secondary outcomes with missing accounting fields. In a fixed-order pilot of 96 matched four-policy blocks, default routing succeeded 17.7 percentage points more often than pinning the cheapest displayed quote (hour-cluster 95% CI 9.2–26.4 pp), but default was always first, so this result is descriptive. A prospective randomized holdout is preregistered and running. The evidence establishes a quote-firmness gap for the pilot workload, not private cost, capacity, market-wide welfare, or strategic manipulation.

1 Introduction

Inference routing is often presented as a model-selection problem: choose between models that differ in price and quality. A marketplace creates a distinct problem. The same model can be supplied by multiple providers, and the displayed provider quote may be conditioned on unobserved capacity, health, fallback, or contractual state. Consequently, the analogy most useful for mechanism design is not an automated market maker. It is the distinction between displayed and delivered RFQ liquidity: a quote can be visible without being a deliverable commitment.

This paper isolates that distinction. It makes six contributions.

1. We formalize a capacity-certified score-routing mechanism. It is the unique entropy-regularized allocation with reliability-weighted inverse-price scores and hard capacity caps.
2. We characterize delivery incentives, a price-only non-guarantee, and a finite-liability impossibility for direct continuous reliability reports.
3. We give a robust linear program for an explicitly declared joint-outage support and prove a max-min delivery-floor comparison to score routing.
4. We introduce a finite-slot collateralized capacity/cost report: an unavailable slot is an expensive sentinel that is never selected truthfully and is costly when falsely reserved. We prove conditional VCG truthfulness for this multidimensional capacity/cost report.

5. We compose that report with an independently audited, finite-grid reliability score. We state the exact funding, verification, and individual-rationality requirements rather than folding them into an unsupported public-price claim.
6. We give a carryover-robust randomized micro-probe design and a value-indexed policy frontier. We characterize the non-identification of fallback and hidden-selection effects from a two-policy comparison and give the minimal randomized intermediate policy that identifies both. We separate its eligibility-conditioned estimand from the all-candidate estimand and give logged-propensity bounds for missing secondary outcomes. A 96-block pilot reveals a large displayed-quote firmness gap; prospective randomizations separate policy from the pilot's fixed-order confound.

The point is deliberately conditional. A certified capacity cap is an assumption, not an inference from a public quote. An audit is independently assigned and objectively verified, not a model of a production router's health signal. These restrictions are essential for interpreting the theorems.

Relation to routing and mechanism design. FrugalGPT, RouteLLM, and RouterBench study model-level cost-quality routing with observed offline outcomes [4, 10, 8]. We instead study provider-level delivery risk for a fixed model. The report incentive component uses the standard separation between allocation and transfers in direct mechanisms [11, 5, 6, 9]. It does not claim that standard VCG alone elicits reliability.

Brown and MacKay show that heterogeneous pricing frequency and algorithmic reaction rules can change price levels and dispersion [2]. Their framework motivates our cadence and reaction-rule screens, but it does not make a displayed provider quote executable. Recent work estimates quality-adjusted prices and markups in the same broad AI service market [12]; our object is instead request-level execution under provider restrictions. Marketplace experiments can be biased by interference [7], while switchback designs require explicit treatment of carryover [1]. Our first-position estimand uses random order to remain valid under unrestricted within-block effects after the first request. Online matching with stochastic rewards and unknown patience studies how a platform should rank limited products for sequential exposure [3]. Our object is complementary: the platform's production provider order is opaque, and the three-arm experiment measures rather than optimizes the separate execution contributions of fallback and delegated selection for the same request. The contemporaneous theory of AI-platform self-preferencing studies routing between an owned model and an outside expert with learning-by-serving [13]; our provider-level mechanism has no vertically integrated model and instead centers latent execution eligibility.

2 Information structure and timing

The mechanism has three deliberately separated stages. At a *contract stage*, the router declares an outside option, a maximum number of collateralizable slots, audit probability, and the source of any score subsidy. These are primitives, not quantities recovered from public prices. At a *report and allocation stage*, providers report a finite capacity/cost curve and, in the audited variant, a reliability value from a clipped finite grid. The router selects reservations using only those reports and the declared rules. At a *delivery and audit stage*, a provider either serves a feasible reservation or incurs a collectable shortfall loss; an independently assigned audit produces a verified Bernoulli outcome.

This timing distinguishes three questions that are often conflated:
1. Can an allocation be feasible under declared hard commitments?
2. Can a provider be induced to deliver a feasible assigned request?
3. Can private capacity, cost, and reliability reports be made incentive compatible?

Object	Used as	Not identified or guaranteed
Posted quote p_i and score q_i	allocation input	marginal cost, selected route, or provider profit
Certified capacity k_i	hard allocation cap	physical service under a real outage
Liability pair (b_i, L_i)	delivery-incentive primitive	enforcement beyond L_i
Audit outcome Y	independent audit signal	a production health measurement process
Public market panel	calibration motivation	welfare, commitment, or realized provider flow

Table 1: Information structure and claim boundary.

The answers use different assumptions and transfers. A shortfall bond can address the second question but does not generally solve the third. A collateralized report can address a restricted capacity/cost problem but does not insure common-mode physical outages. An audit score can address a finite reliability report but requires external funding.

3 Model and claim boundary

Let N be a finite provider set. An epoch has demand $D \geq 0$. Provider $i \in N$ posts a positive request quote $p_i > 0$, has a pre-epoch score $q_i > 0$, a committed capacity $k_i \geq 0$, and marginal delivery cost c_i. The base score is

$$w_i = q_i p_i^{-\eta}, \qquad \eta > 0.$$

Capacity k_i is verified in the base mechanism. It is not a private report and the paper does not claim that a provider reveals it truthfully.

The allocation solves

$$\max_x \left\{ \sum_{i \in N} x_i \log w_i - \sum_{i \in N} x_i \log x_i \right\} \quad \text{s.t.} \quad \sum_i x_i = D, \quad 0 \leq x_i \leq k_i. \tag{1}$$

We use the continuous convention $0 \log 0 = 0$. If aggregate capacity is insufficient, the mechanism assigns all committed capacity and reports residual demand rather than fabricating an assignment.

4 Capacity-certified score routing

4.1 Allocation and delivery

Theorem 1 (Capped score allocation). *If $D \leq \sum_i k_i$, equation* (1) *has a unique allocation x^C and*

$$x_i^C = \min\{k_i, \tau w_i\}, \qquad \sum_i x_i^C = D,$$

for a scalar $\tau \geq 0$. Thus no provider receives more than its certified capacity.

The uncapped analogue assigns Ds_i, where $s_i = w_i / \sum_j w_j$. The distinction matters only when some high-score provider is capacity constrained.

Payments are delivery contingent. A provider that delivers a feasible assigned request receives p_i and pays cost c_i. A deliberate refusal loses a collectable shortfall amount $\min\{b_i, L_i\}$.

Theorem 2 (Delivery incentive)**.** *For a feasible assigned request, delivery rather than deliberate refusal changes provider payoff by*

$$p_i - c_i + \min\{b_i, L_i\}.$$

Delivery is strictly preferred exactly when $\min\{b_i, L_i\} > c_i - p_i$*.*

Theorem 3 (Delivered-count dominance)**.** *Suppose every request assigned within a certified commitment is delivered. If* $D \leq \sum_i k_i$*, capacity-certified routing delivers* D *requests. Uncapped score routing delivers at most*

$$\sum_i \min\{Ds_i, k_i\}.$$

Hence capacity-certified routing weakly dominates uncapped routing in delivered count under these assumptions.

theorem 3 does not compare latency, output quality, total cost, provider utility, or welfare. It is a feasibility comparison only.

4.2 Why price alone is insufficient

Theorem 4 (Price-only separation)**.** *Consider a router that observes only* (p_i, q_i)*, lacks verifiable capacity, does not observe delivery, and cannot enforce a shortfall penalty. No rule that assigns a positive request using only* (p_i, q_i) *can guarantee delivery across all capacity states consistent with those observables.*

The separation is not a claim about any named router. It says that a public price surface cannot certify deliverable liquidity without additional information or enforcement.

5 Robustness to correlated outages

Hard commitments alone do not establish resilience to physical outages. Let Ω^+ be a finite, externally declared support of joint availability states with positive probability. Write $a_i(\omega) \in \{0, 1\}$ for provider i's availability in state ω. No independence assumption is made: one state may remove several providers simultaneously.

The robust allocation solves

$$\begin{aligned} \max_{x,z} \quad & z \\ \text{s.t.} \quad & \sum_i x_i \leq D, \qquad 0 \leq x_i \leq k_i, \\ & \sum_i a_i(\omega) x_i \geq z \qquad \forall \omega \in \Omega^+. \end{aligned} \tag{2}$$

After maximizing z, a secondary objective can use score weights to break ties. Thus price and reliability scores never weaken the max-min delivery guarantee.

Theorem 5 (Robust delivery-floor dominance)**.** *Let* x^S *be any capacity-feasible score allocation and let* x^R *solve equation* (2) *on the same capacities and outage support. Then*

$$\min_{\omega \in \Omega^+} \sum_i a_i(\omega) x_i^R \;\geq\; \min_{\omega \in \Omega^+} \sum_i a_i(\omega) x_i^S.$$

Proposition 1 (Common-mode boundary)**.** *If* Ω^+ *contains a state in which every provider is unavailable, then every non-replicated allocation has robust delivery floor zero.*

theorem 5 is a conditional resilience result, not an estimate of outage probability, expected welfare, or insurance value. Probabilities may be used to report expected delivered count under the same declared law, but they do not enter the max-min objective. proposition 1 is equally important: no allocation rule can recover a positive floor from a support that allows a complete common-mode outage.

6 Private capacity, cost, and reliability reports

6.1 Finite liability boundary

Suppose a provider has true Bernoulli delivery probability q, a positive success margin $m = p - c > 0$, and unbinding physical capacity. If a reliability report $r > q$ increases allocation from $x(q)$ to $x(r)$, write $\ell = \min\{b, L\}$ for the finite collectable loss per failed assignment.

Theorem 6 (Finite liability cannot elicit continuous reliability)**.** *Under payment only for successful service, the expected gain from the over-report is*

$$[x(r) - x(q)]\,[qm - (1 - q)\ell].$$

For every finite ℓ, this gain is positive for $q > \ell/(m+\ell)$. Thus no finite capped shortfall liability makes direct reliability reporting dominant-strategy truthful on a continuous type space where allocation remains report responsive.

The theorem motivates an independent audit. It does not rule out a finite report domain, a non-report-responsive allocation rule, or a different transfer.

6.2 A collateralized finite-slot report

We now replace certified capacity with a restricted private-capacity domain. Provider i has a public maximum of M_i collateralizable reservation slots. Its true report type is a nondecreasing vector

$$t_i = (c_{i1}, \dots, c_{iK_i}, H, \dots, H),$$

where $K_i \le M_i$ is privately known deliverable capacity, each deliverable marginal reservation cost is below a public outside option v, and $H > v$ is a fully collectable shortfall sentinel. A truthful unavailable slot is therefore never selected. If a provider reports an unavailable slot as deliverable and it is reserved, its true cost includes H and the router uses the outside option. Let $\mathcal{T}_i$ denote the finite set of permitted nondecreasing vectors of this form.

For fixed reliability reports r_i, select up to D positive-surplus slots by maximizing reported surplus:

$$\max_x \sum_i \left[r_i v x_i - \sum_{u \le x_i} \hat{c}_{iu} \right], \quad 0 \le x_i \le M_i, \quad \sum_i x_i \le D. \tag{3}$$

The Clarke pivot payment is the externality imposed on other providers and the outside option. This is a reservation-stage construction; the sentinel is collateral, not a claim that a listed capacity limit is physical capacity.

Theorem 7 (Collateralized finite-slot VCG)**.** *Fix reliability reports, v, M_i, and a sentinel $H > v$. On the stated nondecreasing finite report domain, truthful capacity/cost reporting is weakly dominant for the allocation in equation* (3)*. Under the opt-out allocation, truthful utility is nonnegative.*

6.3 Audited finite-grid reliability

Let reliability reports lie in a finite clipped grid $R \subset [\epsilon, 1-\epsilon]$ with $\epsilon > 0$. An audit occurs independently after reports with probability $\rho > 0$; its objectively verified outcome $Y \sim \mathrm{Bern}(q)$ cannot be selected by the provider. On an audit, pay the shifted log score

$$S_A(r, Y) = A[-\log \epsilon + Y \log r + (1-Y)\log(1-r)]. \tag{4}$$

The score is nonnegative and bounded by $A \log((1-\epsilon)/\epsilon)$.

Let $U_i^0(\hat{t}_i, r; q, t_i)$ be provider i's expected utility from the VCG component before the audit score when the true pair is (t_i, q) and report is $(\hat{t}_i, r)$. The conventional opt-out allocation is available.

Theorem 8 (Audited capacity/cost/reliability product report)**.** *Assume the preceding finite domains and independent audit. If, for any $\delta > 0$,*

$$A > \max_{\substack{q,r \in R: q \neq r \\ \hat{t}_i \in \mathcal{T}_i}} \frac{[U_i^0(\hat{t}_i, r; q, t_i) - U_i^0(t_i, q; q, t_i)]_+ + \delta}{\rho \, \mathrm{KL}(\mathrm{Bern}(q) \| \mathrm{Bern}(r))}, \tag{5}$$

then truthful capacity/cost reporting is weakly dominant conditional on a reliability report. The truthful product report is dominant and strictly better than every joint deviation with $r \neq q$. Truthful utility is nonnegative under the VCG opt-out convention.

The finite report domain makes the numerator in equation (5) finite, while clipping makes every distinct Bernoulli KL divergence positive. Therefore a finite A exists. This is not budget balance: a named external fund must reserve at most $nA \log((1-\epsilon)/\epsilon)$ if all n providers are audited. The theorem remains restricted: it does not cover continuous reliability, selectively chosen audits, unfunded collateral, payment-on-success delivery, endogenous capacity investment, or correlated physical failure.

7 Collateral, participation, and funding

The report mechanism is a reservation decision, while delivery still needs collateral and a feasible contract. Let C_i be posted collateral, K_i a known physical capacity in this reduced-form certificate, and $\delta \geq 0$ a required gain from serving a feasible assigned request. With delivery payment p_i and marginal cost c_i, set

$$b_i = \max\{0, \delta - (p_i - c_i)\}, \qquad \bar{k}_i = \begin{cases} \min\{K_i, C_i/b_i\}, & b_i > 0, \\ K_i, & b_i = 0. \end{cases}$$

Allocating at most $\bar{k}_i$ fully collateralizes every shortfall bond.

Proposition 2 (Collateral-feasible delivery certificate)**.** *For an allocation $x_i \leq \bar{k}_i$, locked collateral satisfies $x_i b_i \leq C_i$. If a feasible provider deliberately refuses, serving rather than refusing improves payoff by at least δ per assigned request.*

Proof. For $b_i > 0$, $x_i \leq \bar{k}_i \leq C_i/b_i$ gives $x_i b_i \leq C_i$; the $b_i = 0$ case is immediate. Serving rather than deliberately refusing changes payoff by $p_i - c_i + b_i \geq \delta$. This does not cover physical inability or an uncollectable collateral claim. □

Participation and router funding are separate. Let R_i be a non-contingent reservation transfer, γ_i the provider's opportunity cost per locked collateral dollar, and u_i its all-served outside option. The smallest nonnegative transfer satisfying all-served participation for a positive allocation is

$$R_i^* = \max\{0,\ u_i - x_i[p_i - c_i - \gamma_i b_i]\}.$$

For a two-state router budget, let f_i be fallback cost per missed request and A_i the epoch audit cost. The minimum treasury covering all-served and all-rationed states is

$$T_i^* = \max\{0,\ R_i + x_i p_i + A_i,\ R_i + x_i f_i + A_i - x_i b_i\}. \tag{6}$$

equation (6) is ex-post contract-feasibility accounting. It is not budget balance, revenue, truthful participation, counterparty-solvency, or physical-outage insurance.

8 Algorithms and synthetic theorem illustrations

Capped score allocation. If $D \geq \sum_i k_i$, return every cap and residual $\max\{0, D - \sum_i k_i\}$. Otherwise, sort breakpoints k_i/w_i. Move every binding provider into a capped set B until

$$\tau = \frac{D - \sum_{i \in B} k_i}{\sum_{i \notin B} w_i}$$

satisfies $\tau w_i \leq k_i$ for every uncapped provider. Return k_i on B and τw_i otherwise. Sorting costs $O(|N| \log |N|)$ and the scan is linear.

Robust allocation. equation (2) is a linear program with $|N| + 1$ variables and $|\Omega^+| + 1$ main constraints. A second linear program may maximize the score-weighted allocation subject to retaining the optimal delivery floor. The implementation therefore makes the resilience/price trade-off inspectable rather than hiding it in a scalar reliability score.

Synthetic examples. figure 1 is generated from the repository's deterministic mechanism implementation. In the left two panels, the cheap provider has price 1, the independent provider has price 2, each has capacity 10, and the declared support contains a no-outage state with probability 0.8 and a cheap-provider-outage state with probability 0.2. Score water-filling assigns 8 versus 2 requests; the robust LP assigns all 10 to the independent provider. Its delivery floor is therefore 10 instead of 2. The right panel enumerates finite capacity/cost reports in a separate two-provider example: the false reliability report has strictly positive minimum truthful-payoff advantage at the audit scale prescribed by equation (5). These are illustrations of stated primitives, not fitted empirical results.

9 Hidden eligibility and a value-indexed routing experiment

The separation result in theorem 4 is a worst-case statement. We next give an experiment that measures the value of information hidden behind a router without claiming to observe that information. Let $\mathcal{P}$ be a finite set of routing policies containing delegated default routing d and provider-pinned policies. In block b, let $S_b(p; h) \in \{0, 1\}$ and $C_b(p; h) \geq 0$ denote success and observed billed cost when policy p is run after within-block history h. These are request outcomes, not provider costs.

Draw a random permutation of $\mathcal{P}$ independently in each block. Write P_b for the first policy and $\pi_{bp} = \Pr(P_b = p) > 0$. The first request has empty history. We require non-anticipation—its

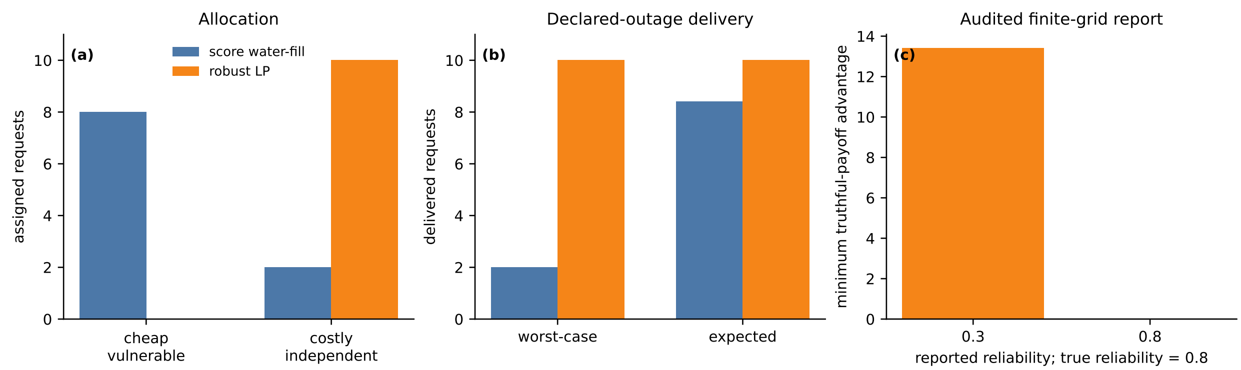

Figure 1: Synthetic theorem illustrations generated by `generate_synthetic_examples.py`. (a) The robust LP shifts allocation away from a cheap provider exposed to a declared outage state. (b) This raises the declared worst-case delivery floor while also improving expected delivery in this particular instance. (c) The finite-grid score makes the false reliability report strictly worse even after minimizing the truthful-payoff advantage over the displayed capacity/cost deviations. No panel is a measurement of a real provider.

potential outcome does not depend on policies that have not yet run—but impose no restriction on the effect of the first request on later requests.

Theorem 9 (First-position identification under post-request interference)**.** *Conditional on the finite collection of first-exposure potential outcomes, the Horvitz–Thompson estimators*

$$\widehat{\mu}_S(p) = \frac{1}{B}\sum_{b=1}^{B} \frac{\mathbf{1}\{P_b = p\}S_b^{\mathrm{obs}}}{\pi_{bp}}, \qquad \widehat{\mu}_C(p) = \frac{1}{B}\sum_{b=1}^{B} \frac{\mathbf{1}\{P_b = p\}C_b^{\mathrm{obs}}}{\pi_{bp}}$$

are unbiased for $B^{-1}\sum_b S_b(p;\varnothing)$ and $B^{-1}\sum_b C_b(p;\varnothing)$. Hence their policy differences identify first-exposure success and observed-spend effects even if any request changes every later potential outcome in its block.

The theorem is deliberately about the first position. A conventional full-block crossover comparison additionally assumes away, models, or bounds carryover. This matters when the likely failure is HTTP 429: sending one probe can change the account-specific state faced by the next probe. Random order turns later positions into a carryover diagnostic rather than silently treating them as independent observations.

Let $\Delta S_p = \mu_S(d) - \mu_S(p)$ and $\Delta C_p = \mu_C(d) - \mu_C(p)$. If a successful response has value $v \geq 0$, the observable policy-utility contrast is

$$\Delta U_p(v) = v\Delta S_p - \Delta C_p. \tag{7}$$

Proposition 3 (Value frontier and delegation rescue)**.** *If $\Delta S_p > 0$, delegated routing is preferred under the request-level value-minus-observed-spend criterion exactly when $v > v_p^* = \Delta C_p/\Delta S_p$. If $\Delta C_p \leq 0$, it weakly dominates for every $v \geq 0$. Moreover,*

$$\Delta S_p = \Pr(S(d) = 1, S(p) = 0) - \Pr(S(d) = 0, S(p) = 1).$$

Thus a positive success effect lower-bounds the frequency with which delegation rescues a pinned failure; under monotone delegation, it equals that frequency.

The default-versus-pinned contrast is itself a composite treatment. Let n pin the cheapest public provider with no fallback, let f submit the full public cheapest-first provider order with fallback, and let d delegate both selection and fallback to the router. Define

$$\Delta_F = \mu_S(f) - \mu_S(n), \qquad \Delta_I = \mu_S(d) - \mu_S(f), \qquad \Delta_T = \mu_S(d) - \mu_S(n).$$

Proposition 4 (Fallback-selection decomposition and non-identification)**.** *The total delegation effect obeys $\Delta_T = \Delta_F + \Delta_I$. A randomized comparison observing only d and n identifies Δ_T but not its two components. In particular, even under the monotonicity restrictions $\mu_S(d) \geq \mu_S(f) \geq \mu_S(n)$, every decomposition $(\Delta_F, \Delta_I) = (a, \Delta_T - a)$ with $a \in [0, \Delta_T]$ is observationally equivalent. Random assignment among d, f, n identifies both components from first-position outcomes under the conditions of theorem 9.*

The proposition changes the interpretation of the experiment. Δ_F is the option value of continuing past the same cheapest first provider under a publicly reproducible order. Δ_I is the incremental value of replacing that order with the router's delegated policy while preserving fallback permission. It may reflect private eligibility, performance scores, contracts, or other undocumented state; the experiment does not identify which signal the router uses. The separately preregistered H81 study randomizes these three policies on ranked models five and six, leaving the H80 four-arm holdout and its stopping rule unchanged.

The randomized support is itself selected. Let $\mathcal{B}$ be the public-rank candidate model-hour blocks and let $E_b = 1$ when the block displays at least two distinct positive-price providers, making all three policies feasible. H81 randomizes only on $\mathcal{B}_E = \{b : E_b = 1\}$. For a secondary outcome $Y_b(p)$, let $R_b(p)$ indicate that its accounting field is observed and suppose a valid pre-treatment support interval $[\ell_b(p), u_b(p)]$ is recorded.

Proposition 5 (Eligibility-conditioned effects and missing-outcome bounds)**.** *First-position randomization identifies $|\mathcal{B}_E|^{-1} \sum_{b \in \mathcal{B}_E} Y_b(p)$, not the corresponding mean over all public-rank candidate blocks. Without assumptions linking eligible and ineligible blocks, the latter effect is not identified from eligible-block outcomes. Within $\mathcal{B}_E$, define*

$$\widehat{\mu}_p^- = \frac{1}{|\mathcal{B}_E|} \sum_{b \in \mathcal{B}_E} \frac{\mathbf{1}\{P_b = p\}}{\pi_{bp}} \left[R_b Y_b + (1 - R_b)\ell_b \right],$$

and define $\widehat{\mu}_p^+$ by replacing ℓ_b with u_b. Their design expectations bound the finite-block mean of $Y_b(p)$, and the contrast between policies p and q lies in

$$[\mu_p^- - \mu_q^+,\ \mu_p^+ - \mu_q^-].$$

If a valid finite upper support is absent in either arm, the corresponding side of the contrast remains unbounded rather than being imputed.

For billed spend, the implementation uses zero as the lower support and a public-quote upper support only for explicit-provider arms whose captured prompt and completion prices cover the fixed prompt (conservatively capped at 64 input tokens) and one output token. Delegated default may leave that public set, so it does not inherit the displayed-set cap. The fixed 60-second HTTP timeout bounds recorded successful-request latency, but latency conditional on success remains a selected secondary description rather than a principal-stratum causal effect. Selected-provider observation and missingness are reported directly by arm.

Equation (7) is a partial welfare object. It avoids choosing a user value by reporting the full frontier, but it omits output-quality differences, provider production cost, router transfers, and externalities. Those omissions prevent a market-wide welfare interpretation.

Model object	Current measurement status	Permitted use in this paper
Posted quote p_i	public quote for a fixed workload	score input and quote heterogeneity
Public performance	exact-match uptime and p90 fields are partial	coverage and missingness audit only
Simulated share $\hat{s}_i$	documented public price proxy	mechanical comparative static only
Selection / retries	716 legacy attempts plus 98 outcome-gated prospective rows	account/workload policy calibration only
Capacity / cost / value	no controlled commitment panel	theorem primitives and protocol fields only
Joint outage support	no joint-outage observations	declared synthetic support only

Table 2: The empirical boundary is part of the mechanism claim.

10 Measurement contract and empirical role

Public provider listings can support descriptive quote and operational heterogeneity, not the premises of the theorems. The companion RouteScope protocol separates public quotes (L0), documented-policy simulations (L1), public operational signals (L2), redacted selected-provider outcomes (L3), and commitment/capacity outcomes (L4).

The public quote simulation uses an inverse-square price rule only as an exposed-policy calibration:

$$\hat{s}_i = \frac{q_i^{-2}}{\sum_j q_j^{-2}}, \qquad \frac{\partial \log \hat{s}_i}{\partial \log q_i} = -2(1 - \hat{s}_i).$$

This derivative is mechanical, not an estimate of demand elasticity. It must not be used as evidence of actual allocation, adverse selection, or MEV-like behavior.

10.1 Brown–MacKay pricing-technology screen

The public quote panel also tests the pricing-technology channel emphasized by Brown and MacKay [2]. The retained completion-price event panel contains 278 changes over 7.53 days. Of 69 providers with sufficient quote exposure, only 19 reprice at all: 10 are classified intraday, seven daily, and two weekly. This supports heterogeneous pricing cadence but leaves most providers in a no-observed-change group.

The stronger reaction-rule implication is not currently estimable. An interim refresh revealed that one additional NextBit change crossed the cadence threshold, moving it from weekly to daily and shrinking the full-panel focal risk set from 15 to three. This exposed outcome look-ahead in classifying a provider with its complete event history. After observing that aggregate instability, we froze cadence classes on the first 70% of events, evaluated only later waves, and removed waves without a complete 24-hour response window; the original classification remains an explicitly outcome-adaptive sensitivity. The corrected holdout has 21 independent waves and 124 rival risk pairs but no slow-initiator/fast-responder exposure, so it supplies no response rate or confidence interval. The full-panel sensitivity has 116 waves and three focal pairs, with no post- or placebo-window move.

A cadence-only regression still finds that slow providers quote a 10.5% premium over fast providers conditional on model. With cadence frozen before the holdout, adding reaction features reduces MAE from 0.1180 to 0.1098 and RMSE from 0.2135 to 0.2119, while the separate strategic hazard model improves log loss but reduces AUC. The reaction exposure and 30-day gates therefore fail. We use pricing technology as a descriptive market characteristic, not evidence of algorithmic reaction, collusion, or Brown–MacKay equilibrium effects.

Policy	Attempts	Success	Wilson 95% CI	HTTP 429
Default router	96	97.9%	[92.7, 99.4]%	1.0%
Pinned cheapest	96	80.2%	[71.1, 86.9]%	18.8%
Pinned second	96	66.7%	[56.8, 75.3]%	33.3%
Pinned random	96	76.0%	[66.6, 83.5]%	21.9%

Table 3: Matched v1 pilot outcomes. Every default request preceded the pinned arms, so intervals describe marginal rates and do not remove the policy/order confound.

10.2 Randomized micro-probe protocol

The fixed-order pilot sent four near-simultaneous one-token requests per hot model: default routing, the cheapest displayed provider, the second-cheapest, and a random remaining displayed provider. Pinned policies disabled fallback. The prospective v2 collector uniformly permutes all four policies, publishes a block identifier, seed, position, and first-position assignment probability, and uses only position zero for the primary contrast. The protocol was frozen after inspecting v1 and before collecting v2. Its first confirmatory cut requires at least 40 first-position assignments per policy and 160 valid blocks; HTTP and network failures remain intention-to-treat outcomes. Before that gate, the analyzer exposes only assignment counts and seed-replay diagnostics. At the gate, it freezes the earliest chronological prefix meeting all four arm counts, so a later download cannot silently replace the first confirmatory cut with a larger sample.

A launch audit found that a legacy cross-study descriptive module had pooled prospective rows by policy label and printed outcome aggregates outside the dedicated gated analyzers. For H81 this disclosed all three policy aggregates from its first two blocks; for H80 it made some full-block prospective information indirectly recoverable from totals pooled with the earlier pilot. We then excluded both prospective study identifiers from every legacy outcome-based route analysis and added cross-analysis isolation tests. No assignment, treatment, exclusion, multiplicity, or stopping rule changed after the disclosure. The incident limits analyst blinding for those launch data but does not change the randomized first-position estimator or deterministic earliest-balanced-prefix rule.

The latest pre-outcome audit contains 16 verified H80 position-zero blocks across four models and four hourly clusters; each model contributes four blocks. H81 contains six verified blocks across two models and three runs. All 16 H80 seed replays and all six H81 seed and treatment-control replays pass. Arm counts are $(5, 2, 2, 7)$ for H80 and $(3, 1, 2)$ for H81, so both analyzers remain below their 40-per-arm gates and release no outcome, confidence interval, or p-value. The H81 candidate-funnel table begins after its first four blocks, so the earlier support remains left-truncated. In the first fully audited run, both rank-five and rank-six candidates were eligible: they displayed 25 and three distinct positive-price providers, respectively. This measures public eligibility for the sampled ranks, not the router's private eligible set.

10.3 A falsifiable mechanism implementation study

A mechanism implementation study must pre-register model-by-epoch policy assignment, commitments, served count, fallback cost, a value-minus-cost proxy, and a latency negative control. The primary estimands are:

1. the selected-provider and fallback calibration of pre-request public features;
2. the delivery-floor and shortfall difference between capacity-certified and baseline routing under observed commitments;

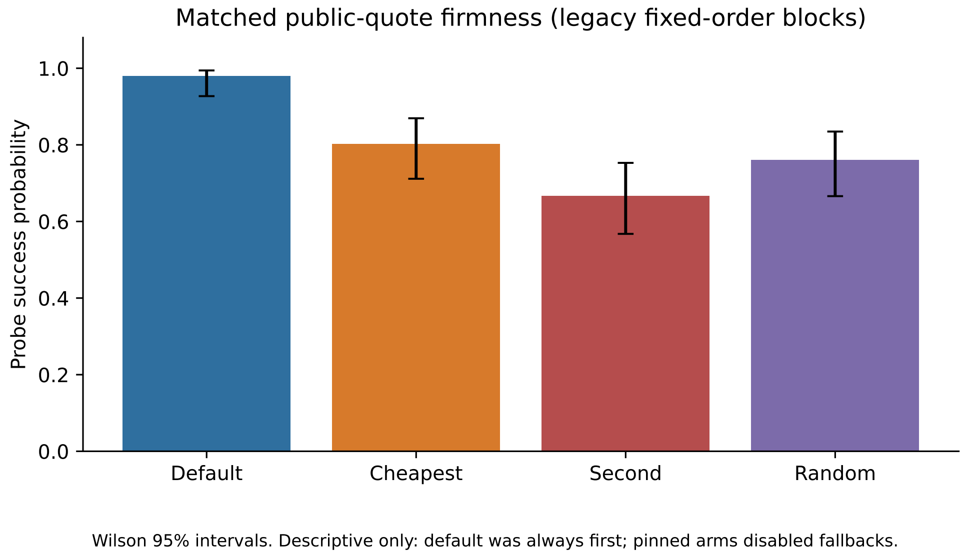

Figure 2: Matched public-quote firmness in the 24-hour v1 pilot. Error bars are Wilson intervals for policy-specific success rates. The prospective v2 first-position analysis, not this figure, supplies the causal contrast.

3. the funded-contract accounting residual under the two polar states in equation (6); and
4. a declared value-net-cost proxy, only when its value and cost inputs are retained and versioned.

Each item has a failure mode. A public shadow can fail to predict selection because live eligibility is hidden. The robust program can fail to improve delivered count if the declared outage support is misspecified. A collateral certificate can be infeasible because a provider cannot fund the required collateral. A welfare claim fails closed whenever request value, delivery cost, or capacity outcome is missing. These are not caveats added after estimation; they are preconditions for interpreting the mechanism.

10.4 Current evidence

The v1 pilot contains 96 complete blocks across four models and 23 hourly common-shock clusters over 24 hours; the median block span is 29.5 seconds. Default routing succeeded 17.7 percentage points more often than the cheapest displayed provider (hour-cluster bootstrap 95% CI 9.2–26.4 pp; cluster sign-flip $p = 0.0019$), 31.3 pp more often than second-cheapest (24.4–38.0 pp), and 21.9 pp more often than random (16.7–27.2 pp). Default selected the displayed cheapest provider in only 21 of 96 blocks and selected a provider outside the three sampled pinned identities in 45 blocks.

For cheapest versus default, the observed-spend difference in the 94 blocks with complete accounting and the 96-block success difference imply $v^* = 2.13 \times 10^{-5}$ dollars per successful request in the pilot. This is a complete-accounting descriptive threshold, not a willingness-to-pay estimate. Most importantly, default was first in every block and pinned requests disabled fallback. The pilot therefore establishes a matched quote-firmness gap for this account and workload, not a causal delegation effect. The preregistered v2 first-position sample is the prospective test.

11 Limitations and conclusion

The mechanism requires more information than a public marketplace currently reveals: verified or collateralized capacity, collectable liability, a finite report domain, independently assigned objective audits, an external audit fund, and a declared joint-outage support. The owned probes identify neither those private inputs nor provider intent, other users' flow, or literal front-running. The current pilot also confounds policy and order. These limitations are the result: displayed prices alone do not supply a delivery guarantee, and the value of hidden eligibility requires a prospective experiment rather than a retrospective causal label.

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A Proofs

A.1 Proof of the capped allocation theorem

Let

$$F = \{x \geq 0 : \sum_i x_i = D,\ x_i \leq k_i\}.$$

The set is nonempty exactly when $D \leq \sum_i k_i$, compact, and convex. The objective in equation (1) is strictly concave on each positive-dimensional feasible face. Its derivative from zero is positive whenever an uncapped positive-weight coordinate is available, so the allocation maximizer is unique. KKT stationarity on an uncapped coordinate gives $\log w_i - \log x_i - 1 = \lambda$, hence $x_i = \tau w_i$. Complementary slackness gives the cap. The function $\sum_i \min\{k_i, \tau w_i\}$ is continuous and nondecreasing from zero to $\sum_i k_i$, so it reaches D. If every coordinate caps, the allocation is still unique even though τ need not be.

A.2 Proofs of delivery and delivered-count results

Delivery yields $p_i - c_i$; deliberate refusal yields $-\min\{b_i, L_i\}$. Their difference proves theorem 2. For theorem 3, the capped allocation sums to D and every assigned unit is delivered by assumption. Under the uncapped rule, provider i can deliver at most $\min\{Ds_i, k_i\}$, and summing gives the result.

A.3 Proof of the price-only separation

Fix a public profile (p_i, q_i) and a provider receiving a positive allocation. Two capacity states are observationally equivalent to the rule: one has enough capacity to serve its allocation; the other has zero capacity. The rule makes the same assignment in both states, while delivery is impossible in the second. Without delivery observation or enforcement, the rule cannot repair that failure from (p_i, q_i) alone.

A.4 Proof of first-position identification and the value frontier

Condition on all first-exposure potential outcomes. Random assignment gives

$$\mathbb{E}\left[\frac{\mathbf{1}\{P_b = p\} S_b^{\mathrm{obs}}}{\pi_{bp}}\right] = \frac{\pi_{bp} S_b(p; \varnothing)}{\pi_{bp}} = S_b(p; \varnothing),$$

where non-anticipation lets the observed first-position outcome equal the corresponding empty-history potential outcome. Summing over blocks and dividing by B proves the success claim; the cost claim is identical. No outcome after the first position appears in the estimator, so its dependence on earlier requests is unrestricted.

For a policy p, expected value-minus-observed-spend is $v\mu_S(p) - \mu_C(p)$. Subtracting pinned from delegated policy gives equation (7). When $\Delta S_p > 0$, division by ΔS_p gives the threshold; if also $\Delta C_p \leq 0$, the difference is nonnegative for every $v \geq 0$. Finally, partition the binary pair $(S(d), S(p))$ into its four possible values. The equal pairs cancel from $\mathbb{E}[S(d) - S(p)]$, leaving rescue probability minus harm probability. Rescue probability is therefore at least $\max\{\Delta S_p, 0\}$, and monotonicity sets the harm term to zero.

For proposition 4, add and subtract $\mu_S(f)$ in Δ_T. If only $\mu_S(d)$ and $\mu_S(n)$ are observed, choose any $a \in [0, \Delta_T]$ and set $\mu_S(f) = \mu_S(n) + a$. This intermediate mean lies between the observed endpoint means and produces $(\Delta_F, \Delta_I) = (a, \Delta_T - a)$ without changing the distribution of either observed arm. Randomizing the intermediate policy makes $\mu_S(f)$ an identified first-position mean by theorem 9, so subtraction identifies both components.

For proposition 5, condition first on the realized eligible set $\mathcal{B}_E$. The same randomization identity used in theorem 9 applies on that fixed set. For every assigned block, the support restriction gives the pointwise inequalities

$$R_b Y_b + (1 - R_b)\ell_b \;\le\; Y_b \;\le\; R_b Y_b + (1 - R_b)u_b.$$

Multiplication by $\mathbf{1}\{P_b = p\}/\pi_{bp}$, summation, and expectation over the randomized assignment proves the arm bounds. The outer contrast combines the lower endpoint for one arm with the upper endpoint for the other. If an upper endpoint is infinite or unavailable, the associated inequality does not yield a finite upper bound. Finally, construct two data-generating processes that agree on the eligibility rule and every potential outcome for $E_b = 1$, but assign different potential outcomes to blocks with $E_b = 0$. They induce the same randomized eligible-block data and different all-candidate effects, proving non-identification without a cross-eligibility restriction.

A.5 Proof of robust delivery-floor dominance

The feasible set of equation (2) is nonempty because $x = 0, z = 0$ is feasible, and it is compact because demand and every capacity cap bound the variables. Take the capacity-feasible score allocation x^S and set

$$z^S = \min_{\omega \in \Omega^+} \sum_i a_i(\omega) x_i^S.$$

Then (x^S, z^S) is feasible for equation (2). Optimality of (x^R, z^R) therefore gives $z^R \ge z^S$, which is the claimed inequality. A secondary score objective ranges only over allocations with $z = z^R$, so it cannot lower the floor. If a support state has $a_i(\omega) = 0$ for all i, every feasible allocation has delivered count zero in that state, proving proposition 1.

A.6 Proof of the finite-liability boundary

At true reliability q, expected payoff per assigned request is $qm - (1 - q)\ell$. Multiplying by the allocation increment $x(r) - x(q)$ gives the displayed gain. The bracket is positive exactly when $q > \ell/(m + \ell)$.

A.7 Proof of collateralized finite-slot VCG

Fix reliability reports. Let W_{-i}^* be the maximum reported surplus without provider i, including the outside option, and let $W_{-i}(x)$ be the other-participant plus outside-option surplus under allocation x. The pivot payment equals the provider's reported own buyer value plus $W_{-i}(x) - W_{-i}^*$. For a true type t_i and report $\hat{t}_i$, expected utility before the audit is a report-independent constant plus the total surplus evaluated with t_i at the allocation selected by $\hat{t}_i$. Truthfully reporting t_i selects an allocation that maximizes that true surplus, so no capacity/cost report can improve utility. The opt-out allocation is feasible; removing a provider weakly restricts achievable surplus, which gives nonnegative truthful utility. A false unavailable slot is included in the true cost at $H > v$, so it cannot evade this comparison.

A.8 Proof of the audited finite-grid construction

Hold reliability reports fixed. The preceding pivot argument makes the true capacity/cost type weakly optimal conditional on those reports. For an audit outcome,

$$\mathbb{E}_q[S_A(q,Y) - S_A(r,Y)] = A\,\mathrm{KL}(\mathrm{Bern}(q)\|\mathrm{Bern}(r)).$$

Because an audit occurs with probability ρ, the expected score advantage is ρA times this quantity. Conditional VCG truthfulness bounds the utility from any joint report $(\hat{t}_i, r)$ by the utility from (t_i, r). When $r \neq q$, equation (5) makes the score advantage exceed the largest remaining VCG gain by δ. A deviation retaining $r = q$ is weakly unprofitable by VCG. The opt-out allocation gives nonnegative VCG utility, and the shifted score is nonnegative, proving individual rationality.